Zero Trust Security with OCI

OUT1015 Practice 0: Prepare Your Summit Retail Environment

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Introduction

About This Workshop

Summit Retail is moving its online store to Oracle Cloud Infrastructure (OCI). Asha leads cloud security for the move. Over one day she handles a series of security incidents, and each one leads to a Zero Trust decision: grant access on purpose, verify every request, and assume that one control can fail.

You watch some incidents as instructor demonstrations, and you fix three of them yourself. Each practice starts with what Asha sees, shows you the part of the architecture involved, and then gives you short tasks. After each task you compare your result with the expected result.

This guide covers **Practice 0: Prepare Your Summit Retail Environment**. The workshop has four practices, each in its own guide. Do Practice 0 first: it builds the environment that the other practices use. Practices 1, 2, and 3 each need only Practice 0, so you can do them in any order.

Practice	What you do	Time
Practice 0	Prepare your Summit Retail environment (Cloud Shell, Terraform)	20 minutes
Practice 1	Get the password out of the code (OCI Vault, instance principals)	25 minutes
Practice 2	Keep the private application private (Security Lists, NSGs, Bastion)	30 minutes
Practice 3	Inspect the requests, not only the port (Web Application Firewall)	35 minutes

Architecture

Practice 0 builds everything in the diagram except the vault, which the lab team prepared, and the WAF policy, which you add in Practice 3. The badges show where each practice works.

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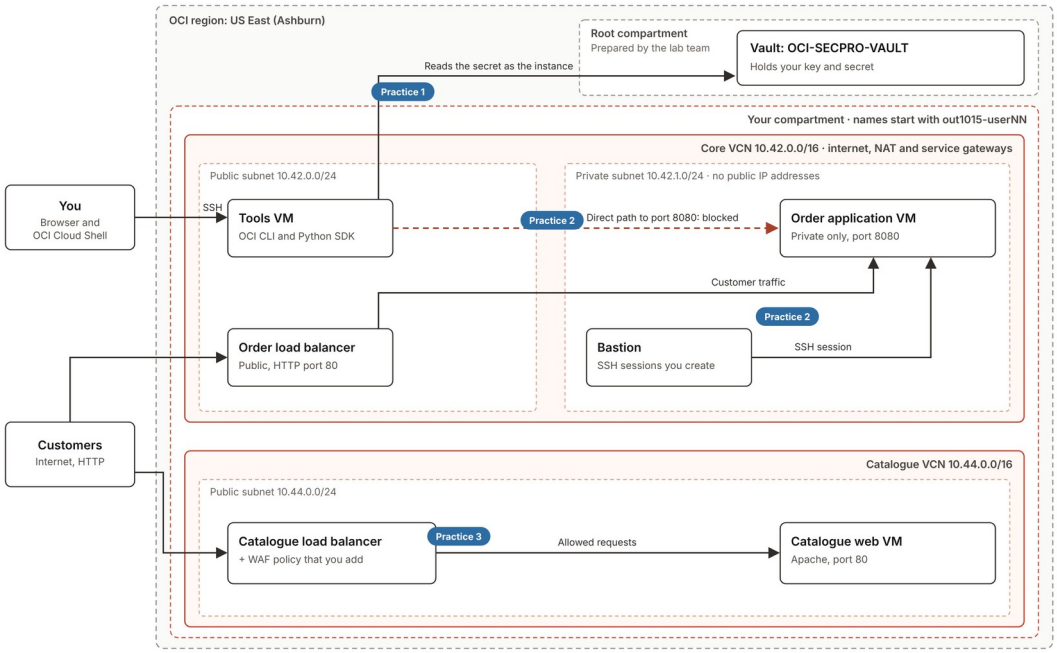


Figure 1: The Summit Retail lab environment in your compartment

The core VCN holds the order system. Customers reach the order application only through the public load balancer. The application server has no public IP address, and you reach it for administration only through OCI Bastion. The Tools VM plays the role of an administrator's or another team's machine inside the same VCN. A second VCN holds the product catalogue, which you protect with WAF.

The Summit Retail Incidents

Incident	What happens	How you take part
1. Identity	A user with read access stops a production instance. Asha finds the grant behind it, adds an explicit deny, and moves sign-in to single sign-on.	Instructor demonstration
2. Secrets	The order application's database password sits in its code. Asha moves it to OCI Vault and lets the instance read it with its own identity.	**Practice 1** (hands-on)
3. Network	The private order application can be reached on a path nobody intended. Asha uses OCI Bastion, finds the broad rule, and removes it.	**Practice 2** (hands-on)
Business path	Zero Trust Packet Routing (ZPR) lets only the intended business path reach the database.	Instructor demonstration
4. Web requests	Attack patterns and floods of requests arrive on the shop's open web port. Asha adds OCI Web Application Firewall (WAF).	**Practice 3** (hands-on)

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Incident	What happens	How you take part
5. Data	A test database holds real customer data. Asha finds it and masks it with Oracle Data Safe.	Instructor demonstration
6. Posture	A storage bucket becomes public. Cloud Guard detects it, and Security Zones prevent it from happening again.	Instructor explanation

Objectives

In the hands-on practices, you:

- Deploy the Summit Retail environment from OCI Cloud Shell with a prepared Terraform package.
- Move a password into OCI Vault and retrieve it at runtime with an instance principal instead of a stored credential.
- Reach a private server through OCI Bastion, then find and remove a network rule that exposes it.
- Detect, block, and rate-limit web requests with OCI Web Application Firewall, and confirm the results in the logs.

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Get Started

Prerequisites

- The lab account details from your instructor: the sign-in URL or tenancy name, your username and password, and your assigned compartment.
- A laptop with a current web browser. You need no other software, because every command runs in OCI Cloud Shell.

Sign In to the Lab Tenancy

Your instructor gives you a lab account in a shared tenancy. Every participant works in their own compartment, so your resources never mix with anyone else's.

Sign In and Check Your Region

1. Open the sign-in URL from your lab details, or go to `https://cloud.oracle.com` and enter the tenancy name.

2. Sign in with the username and password from your lab details. If you are asked to change the password, choose a new one and keep it for the rest of the day.

3. In the Console header, open the region selector and select **US East (Ashburn)**. Every practice uses this region.

4. From the navigation menu, select **Identity & Security**, and then click **Compartments**. Find your assigned compartment in the list, for example `98507690-C07`.

   > **Expected result**
   >
   > You are signed in, the region is US East (Ashburn), and you can see your assigned compartment.

Find Your Participant Number

1. Look at your username. It ends with your participant number, for example `98507690-lab.user07`.

2. Write down the last part, `userNN`, for example `user07`. Everywhere this guide shows `userNN` or `NN`, use your own number.

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> **Note**
>
> Your compartment uses the same number: user07 works in compartment 98507690-C07. The Practice 0 package reads both from your username, and asks you to confirm them.

How to Read This Guide

You see	It means
A light box with a red left edge	Text that you enter or select in the OCI Console.
A dark box that starts with $	A command to run. The task tells you where: in Cloud Shell or on a VM. Do not type the $.
A gray box	Output that you should see. Details such as IDs, dates, and IP addresses differ for you.
Expected result	What to check before you move on.
`<value>`	A placeholder. Replace it with your own value, without the angle brackets.

> **Note**
>
> Console labels sometimes move between releases. If a button is not where the guide says, look for the same action in the **Actions** menu or under the three-dot icon.

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Practice 0: Prepare Your Summit Retail Environment

Introduction

Before Asha can fix anything, Summit Retail's systems must be running. In this practice you build them in your compartment with one prepared Terraform package: two VCNs, three virtual machines, two load balancers, and a bastion. The practices that follow change security controls only, so you build this once.

Estimated time: 20 minutes, most of it waiting while the servers start.

Objectives

- Download the setup package into OCI Cloud Shell.
- Confirm your participant number and compartment, and review the plan before anything is created.
- Deploy the environment and check that both web applications respond.

Practice 0-1: Open Cloud Shell and Download the Package

1. In the Console header, click the **Developer tools** icon and select **Cloud Shell**. Wait until the prompt appears at the bottom of the window.

2. Download the package and list its contents:

```
$ REPO=https://github.com/SaurabhOCI/aiworld-2026-zero-trust-labs
$ ZIP=OUT1015_Participant_Foundation_v6.zip
$ wget -q -O ~/$ZIP $REPO/raw/main/lab0/$ZIP
$ unzip -l ~/$ZIP | tail -1
```

```
    46119                   16 files
```

3. If the last line is not `16 files`, download the same package from Oracle Object Storage instead, and list it again:

```
$ OS=https://objectstorage.us-ashburn-1.oraclecloud.com
$ P=7-3XsOEt1Q48HA2iREfVy1ZpHN94urCBIa
$ P=${P}ySLK6lMScVxL36NTdpqJZ4hdCYCJc9
$ B=n/intoraclerohit/b/out1015-aiworld-2026-labs
$ wget -q -O ~/$ZIP $OS/p/$P/$B/o/lab0/$ZIP
$ unzip -l ~/$ZIP | tail -1
```

```
    46119                   16 files
```

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4. Extract the package into a new folder and move into it:

```
$ unzip -n ~/OUT1015_Participant_Foundation_v6.zip -d ~/out1015-lab
$ cd ~/out1015-lab
```

> **Note**
>
> Keep the `~/out1015-lab` folder for the whole workshop. It holds your SSH key and the record of what the package created, which it needs for clean-up.
>
> If both downloads fail, ask your instructor for the zip file, and upload it with the Cloud Shell menu (**Upload**).

Practice 0-2: Prepare Your Settings

1. Run the preparation step:

```
$ python3 lab.py prepare
```

2. The package shows your username and proposes your participant number and compartment. Check them, and press **Enter** to accept each value. Type a different value only if the proposal is wrong.

```
Signed-in user: 98507690-lab.user07
Participant number (userNN) [user07]:
Your assigned compartment name [98507690-C07]:
```

3. When the package lists the availability domains, press **Enter** to accept availability domain 1.

> **Expected result**
>
> The last lines read:
>
> ```
> SSH private key stays in Cloud Shell: /home/<you>/out1015-
> lab/.local/workshop_ssh
> No infrastructure created. Next: python3 lab.py plan
> ```

> **Note**
>
> The package creates an SSH key pair for you. The private key never leaves Cloud Shell. You use it to connect to the Tools VM in Practice 1, and Bastion uses its public half in Practice 2.

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Practice 0-3: Review the Plan

1. Ask Terraform what it would create. Nothing is created yet:

```
$ python3 lab.py plan
```

2. Scroll through the plan if you like. Every name starts with `out1015-userNN`. Then check the summary at the end:

```
Reviewed plan scope: {"create": 48, "update": 0, "delete": 0}
```

> **Note**
>
> Just above the summary, Terraform suggests running `terraform apply`. Ignore that line, and use the package's own command in Practice 0-4.
>
> The package refuses a plan that would change or delete anything, or touch another compartment. This protects your environment and everyone else's.

Practice 0-4: Deploy the Environment

1. Apply the plan that you reviewed:

```
$ python3 lab.py apply
```

2. Wait for Terraform to finish. It takes about 5 minutes; the load balancers take the longest. The command then checks the servers once:

```
Apply complete! Resources: 48 added, 0 changed, 0 destroyed.
...
Resources created. The servers now finish starting; checking them:
```

> **Note**
>
> Right after `apply`, the check usually reports `not verified` or `Not ready yet`. That is expected: the servers are still installing their software. Continue with Practice 0-5.

> **Caution**
>
> If `apply` stops with a limit, quota, or capacity error, do not run it again in another folder. Keep this folder as it is, and tell your instructor.

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Practice 0-5: Check Your Environment

1. The servers install their software after they start. Check the environment:

```
$ cd ~/out1015-lab
$ python3 lab.py status
```

2. If the last line reads `Not ready yet`, wait a minute and run the command again. The private VM can take up to 15 minutes after `apply` before its Bastion plugin runs. When everything is ready, the output looks like this:

```
Your Summit Retail environment (all values are saved in
.local/outputs.json):
  tools_public_ip  150.136.72.83
  app_private_ip   10.42.1.207
  order_url        http://129.80.254.190
  catalogue_url    http://157.151.217.8
Orders: HTTP 200; expected body matched: True
Catalogue: HTTP 200; expected body matched: True
Bastion plugin on the order application VM: RUNNING
Environment ready.
```

3. Copy the four values at the top into a text file. Your addresses differ from the example; the practices use them:

Value	Used for
`tools_public_ip`	Connecting to the Tools VM with SSH (Practices 1 and 2)
`app_private_ip`	The order application's private address (Practice 2)
`order_url`	The public order site (Practice 2)
`catalogue_url`	The public product catalogue (Practice 3)

4. Open the `catalogue_url` in a new browser tab. The Summit Retail catalogue shows three products.

> **Expected result**
>
> `status` reports **Environment ready.**, and the catalogue page opens in your browser.

Learn More

- OCI Cloud Shell: https://docs.oracle.com/en-us/iaas/Content/API/Concepts/cloudshellintro.htm
- Terraform provider for OCI: https://docs.oracle.com/en-us/iaas/Content/dev/terraform/home.htm

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Need Help?

Raise your hand, and an instructor will come to you. Before that, check the common problems below.

- `Expected exactly one active compartment named ...`: check the compartment name in your lab details, and run `python3 lab.py prepare --compartment <name>`.
- `Inputs already exist`: you already ran `prepare` in this folder. Continue with `python3 lab.py plan`.
- `can't open file ... lab.py`: Cloud Shell restarted in your home folder. Run `cd ~/out1015-lab`, and run the command again.
- `Use a fresh participant Cloud Shell ...`: run the package in OCI Cloud Shell, not on your laptop.
- A limit, quota, or capacity error during `apply`: do not retry in another folder. Keep the folder, and tell your instructor.
- `status` reports `not verified` or `Not ready yet`: the servers are still starting. Wait a minute and run `status` again; it can take up to 15 minutes after `apply`.

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Clean Up

Clean up when your instructor tells you to, normally at the end of the day. The lab resources incur charges while they exist.

> **Note**
>
> Remove the environment last. First follow the Clean Up steps in the Practice 1, 2, and 3 guides for the practices you did. A WAF policy that is still attached stops the catalogue load balancer from being deleted.

Remove the Practice 0 Environment

1. In Cloud Shell, go to your Practice 0 folder and review what would be deleted:

```
$ cd ~/out1015-lab
$ python3 lab.py destroy-plan
```

```
Reviewed plan scope: {"create": 0, "update": 0, "delete": 48}
```

2. Delete the resources. When asked, type your prefix, for example `out1015-user07`:

```
$ python3 lab.py destroy
```

```
No Terraform-managed resources remain in this lab state.
```

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